

WRITTEN HW SOLUTIONS - GRADIENT

1. Want to find $\vec{u}$ such that $\nabla f(P) \cdot \vec{u} = 7$ and $\|\vec{u}\| = 1$ (unit vector)
- $\nabla f = \langle y, x+2y \rangle \Rightarrow \nabla f(P) = \langle 2, 7 \rangle$

Let $\vec{u} = \langle u_1, u_2 \rangle$ then

① becomes: $\langle 2, 7 \rangle \cdot \langle u_1, u_2 \rangle = 7 \Rightarrow 2u_1 + 7u_2 = 7$ (*)

② becomes: $\sqrt{u_1^2 + u_2^2} = 1 \Rightarrow u_1^2 + u_2^2 = 1$ (**)

So we must solve this system of equations for u_1, u_2

(*) $\Rightarrow u_2 = 1 - \frac{2}{7}u_1$ plugging this into (**) (*)

$\Rightarrow u_1^2 + \left(1 - \frac{2}{7}u_1\right)^2 = 1$

$\Rightarrow u_1^2 + 1 - \frac{4}{7}u_1 + \frac{4}{49}u_1^2 = 1$ (Multiply by 49)

$\Rightarrow 49u_1^2 - 28u_1 + 4u_1^2 = 0$

$\Rightarrow 53u_1^2 - 28u_1 = 0$

$\Rightarrow (53u_1 - 28)u_1 = 0 \Rightarrow u_1 = 0 \quad \text{or} \quad u_1 = \frac{28}{53}$

$\downarrow$ (*)

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So $\boxed{\langle 0, 1 \rangle \text{ or } \langle \frac{28}{53}, \frac{45}{53} \rangle}$ $u_2 = 1$ $u_2 = \frac{45}{53}$

2. $\nabla T = \langle 2y, 2x-z, -y \rangle$

$\Rightarrow \nabla T(P) = \langle -2, 2, 1 \rangle \leftarrow$ Direction of greatest increase.

$\Rightarrow \|\nabla T(P)\| = \sqrt{4+4+1} = 3 \leftarrow$ max rate of change in the $\langle -2, 2, 1 \rangle$ direction

Additionally, $-3 \leftarrow$ min rate of change in the $\langle 2, -2, -1 \rangle$ direction.

So, yes!

$$\begin{aligned}
 3. \text{ a. } \nabla(kf) &= \langle (kf)_x, (kf)_y, (kf)_z \rangle \leftarrow \text{Defn of Gradient} \\
 &= \langle kf_x, kf_y, kf_z \rangle \leftarrow \text{pull scalars out of partials} \\
 &= k \langle f_x, f_y, f_z \rangle \leftarrow \text{scalar multiplication of vectors} \\
 &= k \nabla f \leftarrow \text{Defn of Gradient.}
 \end{aligned}$$

$$\begin{aligned}
 \text{b. } \nabla\left(\frac{f}{g}\right) &= \left\langle \left(\frac{f}{g}\right)_x, \left(\frac{f}{g}\right)_y, \left(\frac{f}{g}\right)_z \right\rangle \\
 &= \left\langle \frac{f_x g - f g_x}{g^2}, \frac{f_y g - f g_y}{g^2}, \frac{f_z g - f g_z}{g^2} \right\rangle \\
 &= \frac{1}{g^2} \langle f_x g - f g_x, f_y g - f g_y, f_z g - f g_z \rangle \\
 &= \frac{1}{g^2} \left(\langle f_x g, f_y g, f_z g \rangle - \langle f g_x, f g_y, f g_z \rangle \right) \\
 &= \frac{1}{g^2} \left(g \langle f_x, f_y, f_z \rangle - f \langle g_x, g_y, g_z \rangle \right) \\
 &= \frac{1}{g^2} (g \nabla f - f \nabla g) \\
 &= \frac{g \nabla f - f \nabla g}{g^2}
 \end{aligned}$$